

OHBM 2026 Abstract

Title (100 characters max)

fMRI Decoding Reveals Intact Neural Color Representations in Color Vision Deficiency

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Introduction (2,000 characters max)

Individuals with color vision deficiency (CVD) show profound impairments in red-green color discrimination. However, whether these perceptual deficits reflect a failure of neural color representations in visual cortex remains unclear. Previous neuroimaging studies have reported mixed findings: some suggest reduced neural discriminability in early visual cortex (V1) but preserved responses in higher-level areas (V2-V3),¹ while others propose that color-selective regions like V4 fail to distinguish problematic color pairs in CVD.² It remains unknown whether color discrimination deficits in CVD reflect the absence of color-specific neural representations in early visual cortex. Here, we investigated whether population-level neural color representations differ between individuals with CVD and healthy controls across the visual cortex hierarchy (V1, V2, V3, hV4). We employed a forward encoding model³ to decode color information from fMRI activity patterns, quantifying both classification accuracy and reconstruction precision for eight isoluminant colors presented during a behavioral attention task. If early visual cortex lacks color-specific representations in CVD, decoding accuracy and reconstruction precision should be reduced relative to controls.

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Methods (4,000 characters max)

Under an IRB-approved protocol, we recruited 9 participants: 6 healthy controls (HC; 3 males/3 females, age 22.7±2.5 years) and 3 individuals with CVD (2 deuteranopes and 1 protanomalous individual, 2 males/1 female, age 23.3±2.1 years). CVD diagnosis was confirmed using Ishihara color plates. Functional MRI data were acquired on a 3T Siemens MAGNETOM Trio scanner. We collected T1-weighted MPRAGE structural images (TR=1900 ms, TE=2.52 ms, voxel size=1×1×1 mm³) and T2*-weighted gradient-echo EPI functional images (TR=1500 ms, TE=30 ms, flip angle=75°, voxel size=2×2×2 mm³, 24 oblique slices oriented perpendicular to the calcarine sulcus).

The experimental paradigm was adapted from Brouwer and Heeger (2009).³ Participants viewed 8 isoluminant colors evenly spaced around a circle in CIE Lab* color space (L*=54, radius=38, 45° spacing)

plus a neutral gray (Fig. 1a). Colored circular backgrounds (1.5 s duration) were presented with randomized inter-stimulus intervals of 3–6 s. To control for strategic color processing, participants performed a rapid serial visual presentation (RSVP) task at fixation, detecting transitions from white to black letter 'K' among continuously presented letters (400 ms each). Each participant completed 6 functional runs of approximately 7 minutes each (total scan time ~60 minutes including breaks), with each color presented 48 times total (8 trials per run, Fig. 1b).

Preprocessing was performed using fMRIPrep, including field map-based distortion correction, motion correction, slice-timing correction, and spatial normalization to MNI space (2 mm isotropic). Visual cortex regions of interest (V1, V2, V3, hV4) were defined bilaterally using the Wang et al. (2015) probabilistic atlas.⁴ For each participant and ROI, we estimated voxel-wise response amplitudes (beta coefficients) for each color using a general linear model with motion parameters and drift regressors, followed by high-pass filtering. We selected informative voxels using ANOVA F-tests ($k=1-200$, optimized per subject and ROI using nested cross-validation).

Color reconstruction was formulated as an encoding-decoding pipeline (Fig. 1c), in which stimulus colors were first represented in a low-dimensional channel space and voxel responses were modeled as linear mixtures of these channel responses. We implemented a forward encoding model³ with 6 half-wave rectified squared sinusoidal basis functions (channels) evenly distributed around the color circle. This model assumes that each voxel's response is a weighted sum of these channel responses. For each ROI, we used leave-one-run-out cross-validation to: (1) estimate the linear mapping (weight matrix W) from channel activations to voxel responses using training data, (2) invert this mapping using regularized pseudo-inverse to predict channel responses from held-out voxel data, and (3) reconstruct color angles by selecting the hue whose idealized channel response profile maximized correlation with the predicted channel responses. We quantified decoding performance using: (1) reconstruction error from the forward encoding model (circular distance in degrees between presented and reconstructed colors, random baseline=90°; see Fig. 2a for examples), (2) proportion of reconstructions within $\pm 22.5^\circ$ and $\pm 45^\circ$ of true color, and (3) classification accuracy via diagonal linear discriminant analysis applied directly to voxel response patterns as a complementary metric (8-way classification, chance=12.5%). We compared CVD and healthy control groups using independent-samples t-tests and Cohen's d effect sizes.

To test whether reconstruction reflected genuine color-specific neural representation rather than arbitrary label assignment, we performed permutation testing by selectively shuffling red-green color labels—the color axis most affected in CVD—and re-running reconstruction (Fig. 2c). We tested permutation effects within each group using paired-samples t-tests and Cohen's d effect sizes.

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Results (4,000 characters max)

All three CVD participants demonstrated successful color decoding across the visual hierarchy, with individual performance consistently overlapping the healthy control range. In V1, each CVD participant showed classification accuracies (54.2%, 58.3%, 54.2%) substantially exceeding chance (12.5%) and falling within the healthy control distribution (33.3% to 83.3%). V1 reconstruction errors for CVD participants (40.2°, 39.0°, 48.1°) all fell well below the random baseline (90°) and within the healthy control range (27.4° to 68.0°). This individual-level consistency across all three CVD participants extended to higher visual areas (V2, V3, hV4).

At the group level, healthy controls and CVD participants showed statistically comparable performance across all visual cortex regions. For reconstruction error, the groups showed comparable performance in V1 (Fig. 2b - HC: $46.7 \pm 17.0^\circ$, CVD: $42.4 \pm 4.9^\circ$, $t(7)=0.41$, $p=.694$, $d=-0.29$), V2 (HC: $56.9 \pm 16.8^\circ$, CVD: $55.3 \pm 5.1^\circ$, $t(7)=0.16$, $p=.876$, $d=-0.11$), V3 (HC: $82.8 \pm 14.1^\circ$, CVD: $78.9 \pm 7.5^\circ$, $t(7)=0.43$, $p=.675$, $d=-0.31$), and hV4 (HC: $82.1 \pm 4.6^\circ$, CVD: $76.3 \pm 3.9^\circ$, $t(7)=1.89$, $p=.105$, $d=-1.32$). All regions showed reconstruction errors substantially below the random baseline, with a hierarchical pattern of increasing error from early to higher visual areas ($V1 < V2 < V3 \approx hV4$). In V1 and V2, the majority of reconstructions for both groups fell within $\pm 45^\circ$ of the true color (V1: HC 78%, CVD 81%; V2: HC 62%, CVD 67%), with a substantial proportion within $\pm 22.5^\circ$ (V1: HC 51%, CVD 53%).

Classification accuracy results similarly revealed no group differences. In V1, both groups performed well above chance (HC: $56.6 \pm 18.6\%$, CVD: $55.6 \pm 2.4\%$, $t(7)=0.09$, $p=.930$, $d=-0.06$; chance=12.5%). V2 showed comparable above-chance performance (HC: $43.8 \pm 17.2\%$, CVD: $43.0 \pm 13.9\%$, $t(7)=0.06$, $p=.951$, $d=-0.04$). Higher-level regions V3 (HC: $23.3 \pm 9.1\%$, CVD: $27.8 \pm 8.4\%$, $t(7)=0.72$, $p=.496$, $d=+0.51$) and hV4 (HC: $24.3 \pm 9.5\%$, CVD: $26.4 \pm 9.6\%$, $t(7)=0.30$, $p=.768$, $d=+0.22$) showed modest above-chance accuracies with no group differences. The hierarchical degradation in decoding performance from V1 to hV4 followed the pattern reported in previous forward encoding studies.³ Effect sizes ranged from negligible to medium across comparisons ($|d| < 0.06$ to 0.51).

Permutation testing validated that decoding was driven by neural color representations. Shuffling red-green color labels during training degraded reconstruction accuracy overall ($d=0.48$), with significant effects in healthy controls ($p=.041$, $d=0.44$) but not in CVD participants ($p=.497$, $d=0.20$, Fig. 2d). However, the small CVD sample size ($n=3$) limits interpretation of this null result.

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Conclusions (4,000 characters max, typically shorter)

Despite profound behavioral deficits in red-green color discrimination, individuals with CVD demonstrate population-level neural color representations in early and intermediate visual cortex (V1 through hV4) that are indistinguishable from healthy controls at both individual and group levels. This dissociation between neural representation and behavioral performance demonstrates that color discrimination impairments in CVD do not arise from failures of color coding in early and intermediate visual cortex. Control analyses indicate that decoding performance cannot be explained by arbitrary label structure or model bias. These results constrain models of color perception by ruling out early visual cortex as the locus of red-green discrimination failure in CVD, and demonstrate that preserved sensory signals are not sufficient to support behavioral color discrimination. This neural-behavioral dissociation advances understanding of the relationship between sensory coding and behavioral color discrimination in the visual system.

Character count: 1,027

References (Maximum 5, AMA style)

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